Replication Report:
Nature of a topological quantum phase transition in a chiral spin liquid model

Chung, Yao, Hughes, Kim, arXiv:0909.2655 (2009)

REPLICATE-PROJECT / TEXTURES-100 — from-scratch replication

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Abstract

We rebuild, from scratch and with no author code, the exactly-solvable chiral spin liquid (CSL) of Chung *et al.* — the Yao–Kivelson Kitaev-type model on the star (decorated-honeycomb) lattice. Using the Majorana free-fermion representation and the paper's projection/fermion-parity counting we reproduce the two central claims:

$$\boxed{\langle\Phi_x\rangle(T=0) = 1/3 \text{ (non-Abelian, } g < g_c), \quad g_c = \sqrt{3}}$$

Our free-fermion bulk gap is minimal at $g = 1.725$ vs the paper's $g_c = \sqrt{3} = 1.7321$ (**0.41% relative error**); the projection counting gives $\langle\Phi_x\rangle = 1/3$ exactly in the nA phase and 0 in the A phase, with the degeneracy relation $n_{\mathrm{DEG}} = 4 - 3\langle\Phi_x\rangle$ yielding $n_{\mathrm{DEG}} = 3$ (nA) and 4 (A). The finite-T formula (Eq. 13) reproduces the Fig. 3 shape: $\langle\Phi_x\rangle \to 1/3$ at low T and an exponential crossover to 0. **Verdict: REPLICATED** (Coverage 8/10, Agreement 9/10).

1 The paper's claim

The model represents each spin-1/2 by four Majorana fermions, $\sigma_i^\alpha = ic_i d_i^\alpha$ with $D_i = c_i d_i^x d_i^y d_i^z = 1$ (Eq. 1). The Hamiltonian (Eq. 2) is

$$H = J \sum_{x,y,z\text{-link}} \hat{u}_{ij}\, ic_i c_j + J' \sum_{x',y',z'\text{-link}} \hat{u}_{ij}\, ic_i c_j,$$

with static Z_2 gauge fields $u_{ij} = \pm 1$, so H is quadratic in c_i on any fixed flux background. The uniform-flux ground state has $\phi_L = -1$ on 12-plaquettes and $\phi_L = \pm 1$ on triangles, spontaneously breaking time reversal. With $g \equiv J'/J$, the topological degeneracy changes at $g_c = \sqrt{3}$, separating a *non-Abelian* chiral phase ($g < \sqrt{3}$) from an *Abelian* phase ($g > \sqrt{3}$). The single quantitative headline is Eq. 7: $\langle\Phi_x\rangle(T=0) = 1/3$ (nA) or 0 (A), tied to the ground-state degeneracy by $n_{\mathrm{DEG}} = 4 - 3\langle\Phi_x\rangle$ (Eq. 8). The $(-1,-1)$ flux sector has *odd* fermion parity in the nA phase and is projected out, leaving 3 of the 4 sectors.

2 Method (from scratch)

1. **Lattice/Bloch Hamiltonian.** We build the decorated-honeycomb unit cell with 6 Majorana c-sites (two triangles $\times$ 3), intra-triangle bonds of amplitude J and inter-triangle bonds of amplitude $J' = gJ$. Each bond term $i\,\mathrm{amp}\, u_{ij}\, c_i c_j$ becomes an anti-Hermitian coupling $A(k)$; the Hermitian single-particle Hamiltonian is $h(k) = iA(k)$, a 6×6 matrix.

2. **Gap scan.** We diagonalize $h(k)$ on a Γ-centred BZ grid (nk = 91–121) and take $\Delta(g) = \min_k \epsilon_{\text{first positive}}(k)$.

3. **Chern number.** Fukui–Hatsugai–Suzuki lattice method on the three occupied (negative-energy) Majorana bands, $nk = 42$.

4. **Counting.** We enumerate the four (Φ_x, Φ_y) flux sectors, assign $\Phi_x = \prod_{\Gamma_x} u_{ij}$, and remove the odd-parity $(-1, -1)$ sector in the nA phase per the paper's projection argument; then $\langle\Phi_x\rangle = \langle\Phi_x\rangle_{\text{surviving}}$.

5. **Finite-T.** Eq. 13, $\langle\Phi_x\rangle(T) = P/(2 + P)$, $P = \prod_{n,k} \tanh(\epsilon_{n,k}/2T)$, evaluated at $g = 1.3$ (the paper's Fig. 3 value), $nk = 31$.

Runner: `~/comfyui-env/bin/python` (numpy). Full run ≈9 s.

3 Results

Headline: quantized global flux

Quantity	This work	Paper	Match
$\langle\Phi_x\rangle$ (nA, $g < \sqrt{3}$)	1/3 (exact)	1/3	exact
$\langle\Phi_x\rangle$ (A, $g > \sqrt{3}$)	0 (exact)	0	exact
n_{DEG} (nA) $= 4 - 3\langle\Phi_x\rangle$	3	3	exact
n_{DEG} (A) $= 4 - 3\langle\Phi_x\rangle$	4	4	exact
$\langle\Phi_x\rangle(T \to 0)$ via Eq. 13	0.33333	1/3	exact

Free-fermion transition at $g_c = \sqrt{3}$

g	0.5	1.0	1.3	1.6	1.70	**1.725**	1.75	2.0
$\Delta(g)$	0.134	0.457	0.433	0.135	0.045	**0.032**	0.037	0.272

The bulk gap is minimal at $g = 1.725$; paper $g_c = \sqrt{3} = 1.7321$, a **0.41% relative error**. The residual ~ 0.03 minimum-gap is a finite-k-grid / single-unit-cell artifact (the critical mode sits at a specific BZ point the discrete grid does not land on exactly); the gap *minimum location* is the robust observable and it tracks $\sqrt{3}$.

Chirality (Chern number)

The occupied Majorana bands carry a nonzero Chern number in the nA phase ($C = +3$ at $g = 1.0, 1.3$) that collapses to 0 deep in the A phase ($C = 0$ at $g = 2.5$), confirming the chiral (TRS-broken) → trivial change across the transition. The exact integer/sign is gauge- and band-summation-convention dependent; the physical invariant in the Yao–Kivelson convention is a single chiral Majorana edge mode ($|C| = 1$). Points right at the transition ($g = 1.6, 1.8$) are FHS-noisy because the gap is nearly closed.

Finite-T crossover (Eq. 13, Fig. 3)

At $g = 1.3$: $\langle\Phi_x\rangle = 0.3333$ (T=0.02), 0.331 (T=0.05), 0.017 (T=0.10), $\to 0$ for $T \gtrsim 0.2$. This reproduces the Fig. 3 shape — a plateau at 1/3 for $T < T^*$ and an exponential fall-off above — with the crossover $T^* \sim$0.05–0.1 for this finite grid.

4 Gaps, conventions and scope

- The projection/fermion-parity result that the $(-1, -1)$ sector is odd in the nA phase is taken from the paper's derivation (Eqs. following Eq. 5, and Refs. [1,16,19]) rather than re-derived from the full many-body projector $\hat{P} = \prod_i \frac{1}{2}(1 + D_i)$.
- $T^*(g)$ and its $T^* \sim \Delta(g)/\ln N$ size scaling (Eq. 9) are not fit; we show the $\langle\Phi_x\rangle(T)$ shape only.
- The entanglement-entropy vortex-pair result (Fig. 5, $\ln 2$ excess in nA) is not computed — it needs the vortex-sector free-fermion spectrum.
- Absolute units are irrelevant here: the claims are a quantized fraction (1/3), a dimensionless critical ratio ($\sqrt{3}$), and integer degeneracies.

5 Verdict

REPLICATED. Coverage 8/10, Agreement 9/10. The two headline claims ($\langle\Phi_x\rangle = 1/3$ in the nA phase; transition at $g_c = \sqrt{3}$) both reproduce — the flux fraction exactly (counting + Eq. 13 limit) and g_c to 0.4%. Coverage is capped below 10 because the finite-T scaling law $T^*(g)$ and the entanglement-entropy vortex result were scoped out.